Vector Calculus and Complex Analysis Midterm I

1. Consider the vector function

$$\boldsymbol{F}(x,y,z)=\left(\frac{-2xz}{(x^2+y^2-1)^2+z^2},\frac{-2yz}{(x^2+y^2-1)^2+z^2},\frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2}\right)$$

defined on $\boldsymbol{R}^3\setminus\{(x,y,z):x^2+y^2=1 \text{ and } z=0\}$.

(a) Show that $\operatorname{curl}(\boldsymbol{F})=\boldsymbol{0}$.

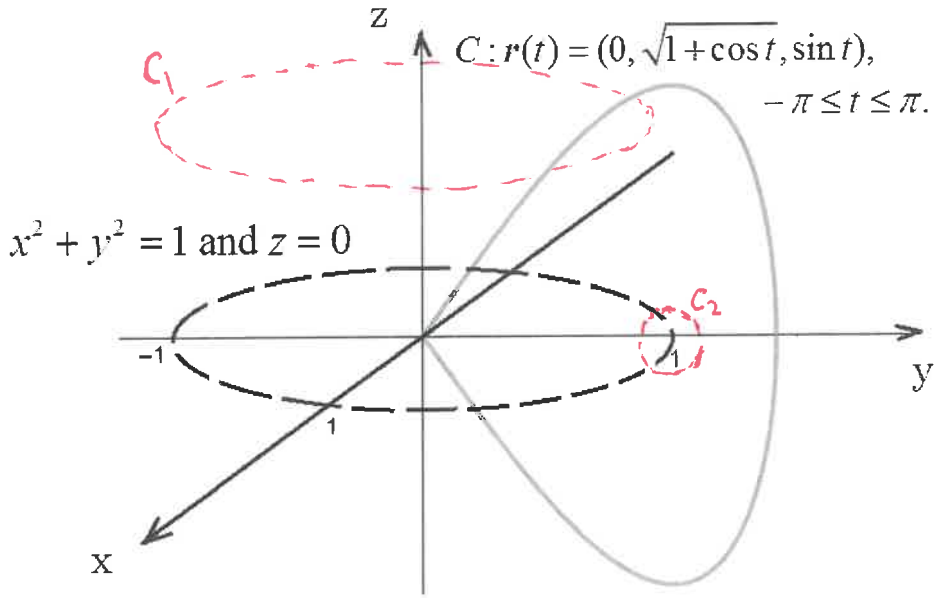
(b) Show that $\boldsymbol{F}(x,y,z)=\nabla\left(\tan^{-1}\frac{z}{x^2+y^2-1}\right)$

(c) Consider the closed curve $C:\boldsymbol{r}(t)=(0,\sqrt{1+\cos t},\sin t),\quad -\pi\le t\le\pi$. Calculate $\int_C \boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}$

(d) Is it possible to find a smooth oriented surface S with boundary C so that $\boldsymbol{F}$ is defined on S?

(e) Calculate $\int_{C_1}\boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}$ and $\int_{C_2}\boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}$ where $C_1:\boldsymbol{r}(t)=(\cos t,\sin t,2),\quad 0\le t\le 2\pi$ and $C_2:\boldsymbol{r}(t)=(0,1+0.1\cos t,0.1\sin t),\quad 0\le t\le 2\pi$

(f) Is $\boldsymbol{R}^3\setminus\{(x,y,z):x^2+y^2=1 \text{ and } z=0\}$ simply connected? Why?



Solution

(a) $\operatorname{curl}(\boldsymbol{F}) = \left(\frac{\partial}{\partial y} \frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2} - \frac{\partial}{\partial z} \frac{-2yz}{(x^2+y^2-1)^2+z^2} \right)\mathbf{i}$

$+\left(\frac{\partial}{\partial z} \frac{-2xz}{(x^2+y^2-1)^2+z^2} - \frac{\partial}{\partial x} \frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2} \right)\mathbf{j} + \left(\frac{\partial}{\partial x} \frac{-2yz}{(x^2+y^2-1)^2+z^2} - \frac{\partial}{\partial y} \frac{-2xz}{(x^2+y^2-1)^2+z^2} \right)\mathbf{k}$

$\frac{\partial}{\partial y} \frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2} = \frac{\left((x^2+y^2-1)^2+z^2\right)2y - (x^2+y^2-1)2(x^2+y^2-1)2y}{\left((x^2+y^2-1)^2+z^2\right)^2} = \frac{2yz^2 - 2y(x^2+y^2-1)^2}{\left((x^2+y^2-1)^2+z^2\right)^2}$

$\frac{\partial}{\partial z} \frac{-2yz}{(x^2+y^2-1)^2+z^2} = \frac{-\left((x^2+y^2-1)^2+z^2\right)2y + 2yz(2z)}{\left((x^2+y^2-1)^2+z^2\right)^2} = \frac{2yz^2 - 2y(x^2+y^2-1)^2}{\left((x^2+y^2-1)^2+z^2\right)^2}$

$\frac{\partial}{\partial z} \frac{-2xz}{(x^2+y^2-1)^2+z^2} = \frac{2xz^2 - 2x(x^2+y^2-1)^2}{\left((x^2+y^2-1)^2+z^2\right)^2} = \frac{\partial}{\partial x} \frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2}$ $\left[\text{by } \frac{\partial}{\partial y} \frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2} \right]$

$\frac{\partial}{\partial x} \frac{-2yz}{(x^2+y^2-1)^2+z^2} = \frac{2yz(2(x^2+y^2-1)2x)}{\left((x^2+y^2-1)^2+z^2\right)^2} = \frac{8(x^2+y^2-1)xyz}{\left((x^2+y^2-1)^2+z^2\right)^2}$

$\frac{\partial}{\partial y} \frac{-2xz}{(x^2+y^2-1)^2+z^2} = \frac{8(x^2+y^2-1)xyz}{\left((x^2+y^2-1)^2+z^2\right)^2}$

$\operatorname{curl}(\boldsymbol{F}) = \mathbf{0}$ (1)

(b) $\frac{\partial}{\partial x} \tan^{-1} \frac{z}{x^2+y^2-1} = \frac{1}{1+(\frac{z}{x^2+y^2-1})^2} \cdot \frac{-2xz}{(x^2+y^2-1)^2} = \frac{-2xz}{(x^2+y^2-1)^2+z^2}$

$\frac{\partial}{\partial y} \tan^{-1} \frac{z}{x^2+y^2-1} = \frac{-2yz}{(x^2+y^2-1)^2+z^2}$

$\frac{\partial}{\partial z} \tan^{-1} \frac{z}{x^2+y^2-1} = \frac{1}{1+(\frac{z}{x^2+y^2-1})^2} \cdot \frac{1}{(x^2+y^2-1)} = \frac{x^2+y^2-1}{(x^2+y^2-1)^2+z^2}$ (1)

(c) $C : \boldsymbol{r}(t) = (0, \sqrt{1+\cos t}, \sin t), \quad -\pi \le t \le \pi.$

$\boldsymbol{F}(\boldsymbol{r}(t)) = \boldsymbol{F}(x,y,z) = \left(0, \frac{-2\sqrt{1+\cos t}\sin t}{(1+\cos t-1)^2+\sin^2 t}, \frac{1+\cos t-1}{(1+\cos t-1)^2+\sin^2 t} \right) = (0, -2\sqrt{1+\cos t}\sin t, \cos t)$ (1)

$\boldsymbol{r}'(t) = \left(0, \frac{-\sin t}{2\sqrt{1+\cos t}}, \cos t \right)$ (1)

$\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r} = \int_{-\pi}^{\pi} (0, -2\sqrt{1+\cos t}\sin t, \cos t) \cdot \left(0, \frac{-\sin t}{2\sqrt{1+\cos t}}, \cos t \right) dt = \int_{-\pi}^{\pi} 1 \, dt = 2\pi$ (1)

(d) By (b), $\operatorname{curl}(\boldsymbol{F}) = \operatorname{curl}\left(\nabla \left(\tan^{-1} \frac{z}{x^2+y^2-1} \right) \right) = 0$. Suppose there exists such surfaces. By Stokes's Theorem, (1)

$$0 = \iint_S \text{curl}(\boldsymbol{F}) \cdot n dA = \iint_R \text{curl}(\boldsymbol{F}) \cdot N du dv = \int_C \boldsymbol{F}(r(s)) \cdot r'(s) ds = 2\pi$$

Contradiction. It is impossible.

(e) By Stokes's Theorem, choose $S_1 = \{(x, y, 2) : x^2 + y^2 \le 1\}$.

$$\int_{C_1} \boldsymbol{F}(r(s)) \cdot r'(s) ds = \iint_{S_1} \text{curl}(\boldsymbol{F}) \cdot n dA = 0$$

There is another surface S_2 with boundary C and $-C_2$. By Stokes's Theorem again,

$$\int_C \boldsymbol{F}(r(s)) \cdot r'(s) ds - \int_{C_2} \boldsymbol{F}(r(s)) \cdot r'(s) ds = \iint_{S_2} \text{curl}(\boldsymbol{F}) \cdot n dA = 0$$

$$\int_{C_2} \boldsymbol{F}(r(s)) \cdot r'(s) ds = \int_C \boldsymbol{F}(r(s)) \cdot r'(s) ds = 2\pi$$

(f) $\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r} \neq 0$ by (c) and $\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r}$ is path dependent in $\boldsymbol{R}^3 \setminus \{(x, y, z) : x^2 + y^2 = 1 \text{ and } z = 0\}$. Therefore $\boldsymbol{R}^3 \setminus \{(x, y, z) : x^2 + y^2 = 1 \text{ and } z = 0\}$ is NOT simply connected.

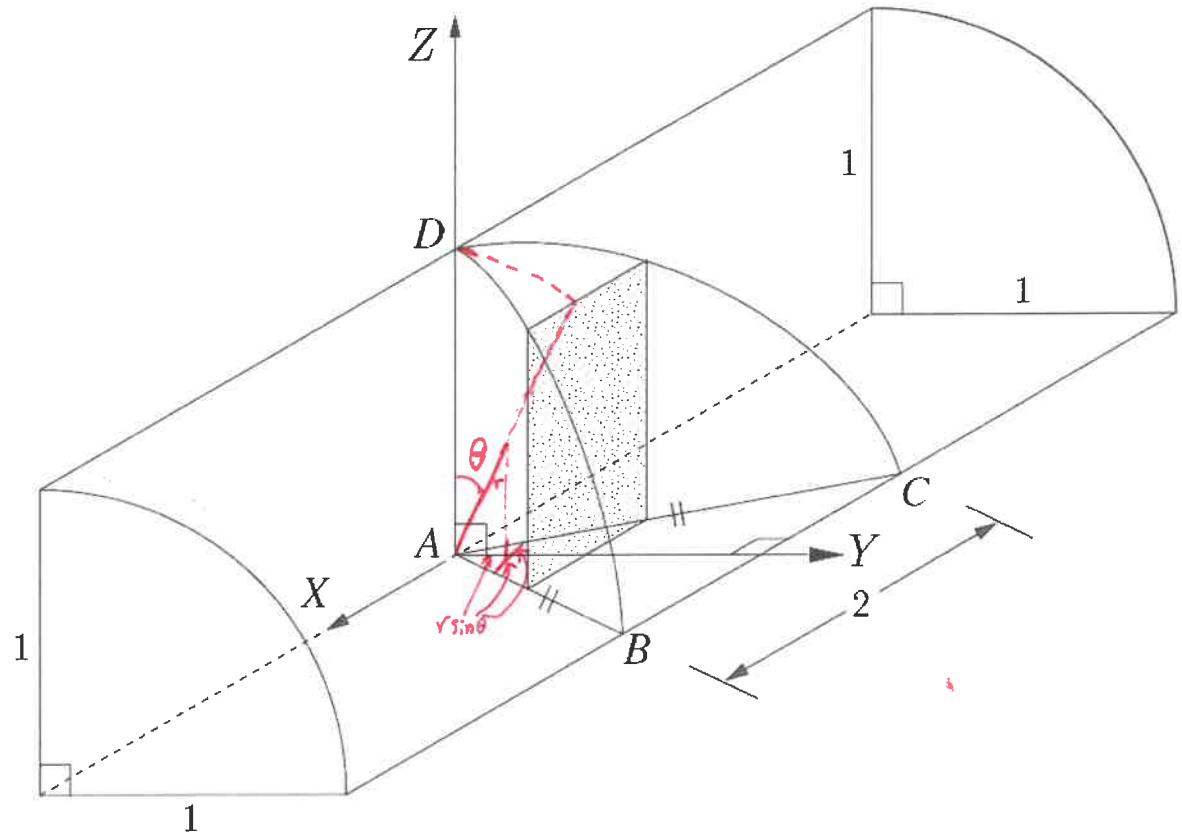
2. The solid $ABCD$ is cut from a quarter cylinder of radius 1 as shown. Its base is an isosceles triangle ABC with $AB = AC$. The length of BC is 2.

The cross-sections perpendicular to Y-axis are rectangles. A typical cross-section is shown shaded in the diagram.

Let S be the curved surface of BCD defined by

$$r(u,v) = (u, v, \sqrt{1-v^2}), \quad -v \le u \le v, 0 \le v \le 1$$

and C the positive oriented boundary of S. Let $\boldsymbol{F}(x,y,z) = \langle x^2, z^2, y^2 \rangle$ be a vector field.



(a) Verify Stokes's Theorem, $\iint_S \text{curl}(\boldsymbol{F}) \cdot n dA = \int_C F(r(s)) \cdot r'(s) ds$.

Hint: $BD : r(s) = (\cos s, \cos s, \sin s) \quad 0 \le s \le \frac{\pi}{2}$

$CD : r(s) = (-\cos s, \cos s, \sin s) \quad 0 \le s \le \frac{\pi}{2}$

$CB : r(s) = (s, 1, 0) \quad -1 \le s \le 1$

(b) Calculate $\iiint_{ABCD} \text{div}(\boldsymbol{F}) dV$.

(a) $r(u,v)=(u,v,\sqrt{1-v^2}),\quad -v\le u\le v,\ 0\le v\le 1$

$r_u=(1,0,0)$

$$\text{curl}(\boldsymbol{F})=\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & z^2 & y^2 \end{vmatrix}=(2y-2z,0,0)$$ (1)

$N\perp r_u \Rightarrow N\perp \text{curl}(\boldsymbol{F})$ (1)

$$\iint_S \text{curl}(\boldsymbol{F})\cdot n dA=\iint_R \text{curl}(\boldsymbol{F})\cdot N du dv=0$$ (1)

$BD: r(s)=(\cos s,\cos s,\sin s)\quad 0\le s\le \frac{\pi}{2}$

$CD: r(s)=(-\cos s,\cos s,\sin s)\quad 0\le s\le \frac{\pi}{2}$

$CB: r(s)=(s,1,0)\quad -1\le s\le 1$

$\boldsymbol{F}(x,y,z)=\langle x^2,z^2,y^2\rangle$

$$\int_{BD} F(r(s))\cdot r'(s)ds=\int_0^{\pi/2}(\cos^2 s,\sin^2 s,\cos^2 s)\cdot(-\sin s,-\sin s,\cos s)ds$$
$$=\int_0^{\pi/2}(-\sin s\cos^2 s-\sin^3 s+\cos^3 s)ds$$ (1)

$$\int_{CD} F(r(s))\cdot r'(s)ds=\int_0^{\pi/2}(\cos^2 s,\sin^2 s,\cos^2 s)\cdot(\sin s,-\sin s,\cos s)ds$$
$$=\int_0^{\pi/2}(\sin s\cos^2 s-\sin^3 s+\cos^3 s)ds$$ (1)

$$\int_{CD} F(r(s))\cdot r'(s)ds-\int_{BD} F(r(s))\cdot r'(s)ds=2\int_0^{\pi/2}\sin s\cos^2 s ds=-\left[\frac{2\cos^3 s}{3}\right]_0^{\pi/2}=\frac{2}{3}$$ (1)

$$\int_{CB} F(r(s))\cdot r'(s)ds=\int_{-1}^{1}(s^2,0,1)\cdot(1,0,0)ds=\int_{-1}^{1}s^2 ds=\frac{2}{3}$$ (1)

$$\iint_S \text{curl}(\boldsymbol{F})\cdot n dA=0=\int_{BC} F(r(s))\cdot r'(s)ds+\int_{CD} F(r(s))\cdot r'(s)ds+\int_{DB} F(r(s))\cdot r'(s)ds=\int_C F(r(s))\cdot r'(s)ds.$$

(b) $\text{div}(\boldsymbol{F})=2x$ (1)

$$\iiint_{ABCD}\text{div}(\boldsymbol{F})dV=\int_0^{\pi/2}\int_0^1\left(\int_{-r\sin\theta}^{r\sin\theta}2x dx\right)r dr d\theta$$ (2)

$$=\int_0^{\pi/2}\int_0^1\left[x^2\right]_{-r\sin\theta}^{r\sin\theta} r dr d\theta$$ (1)

$$=0$$ (1)

ABD: $r(u,v)=(u\cos v,u\cos v,u\sin v)\quad 0\le u\le 1,0\le v\le\frac{\pi}{2}$

$r_u=(\cos v,\cos v,\sin v)$

$r_v=(-u\sin v,-u\sin v,u\cos v)$

outer normal $N=(u,-u,0)$

$$\iint_{ABD} \boldsymbol{F}\cdot n dA = \int_0^{\frac{\pi}{2}}\int_0^1 \boldsymbol{F}\cdot N du dv = \int_0^{\frac{\pi}{2}}\int_0^1 \left\langle u^2\cos^2 v, u^2\sin^2 v, u^2\cos^2 v\right\rangle \cdot \left\langle u,-u,0\right\rangle du dv$$
$$= \int_0^1 u^3 du \int_0^{\frac{\pi}{2}} \cos^2 v - \sin^2 v dv = \frac{1}{4}\int_0^{\frac{\pi}{2}} \cos 2v dv = 0$$

ACD: $r(u,v) = (-u\cos v, u\cos v, u\sin v) \quad 0\le u\le 1, 0\le v\le \frac{\pi}{2}$

$r_u = (-\cos v, \cos v, \sin v)$
$r_v = (u\sin v, -u\sin v, u\cos v)$
outer normal $N = (-u,-u,0)$

$$\iint_{ACD} \boldsymbol{F}\cdot n dA = \int_0^{\frac{\pi}{2}}\int_0^1 \boldsymbol{F}\cdot N du dv = \int_0^{\frac{\pi}{2}}\int_0^1 \left\langle u^2\cos^2 v, u^2\sin^2 v, u^2\cos^2 v\right\rangle \cdot \left\langle -u,-u,0\right\rangle du dv$$
$$= -\int_0^1 u^3 du \int_0^{\frac{\pi}{2}} \cos^2 v + \sin^2 v dv = -\frac{\pi}{8}$$

BCD: $r(u,v) = (u, v, \sqrt{1-v^2}), \quad -v\le u\le v, 0\le v\le 1$

$r_u = (1,0,0)$
$r_v = \left(0, 1, \dfrac{-v}{\sqrt{1-v^2}}\right)$
outer normal $N = \left(0, \dfrac{v}{\sqrt{1-v^2}}, 1\right)$

$$\iint_{BCD} \boldsymbol{F}\cdot n dA = \int_0^1\int_{-v}^{v} \boldsymbol{F}\cdot N du dv = \int_0^1\int_{-v}^{v} \left\langle u^2, 1-v^2, v^2\right\rangle \cdot \left\langle 0, \tfrac{v}{\sqrt{1-v^2}}, 1\right\rangle du dv$$
$$= \int_0^1\int_{-v}^{v} v\sqrt{1-v^2} + v^2 du dv = \int_0^1 2v^2\sqrt{1-v^2} + 2v^3 dv = \frac{1}{2} + 2\int_0^{\frac{\pi}{2}} \sin^2 x\cos^2 x dx = \frac{1}{2} + \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin^2 2x dx$$
$$= \frac{1}{2} + \frac{1}{4}\int_0^{\frac{\pi}{2}} 1-\cos 4x dx = \frac{1}{2} + \frac{\pi}{8}$$

ABC: $r(u,v) = (u,v,0) \quad -v\le u\le v, 0\le v\le 1$

outer normal $N = (0,0,-1)$

$$\iint_{ABC} \boldsymbol{F}\cdot n dA = \int_0^1\int_{-v}^{v} \boldsymbol{F}\cdot N du dv = \int_0^1\int_{-v}^{v} \left\langle u^2, 0, v^2\right\rangle \cdot \left\langle 0,0,-1\right\rangle du dv = \int_0^1\int_{-v}^{v} -v^2 du dv = -\int_0^1 2v^3 dv = -\frac{1}{2}$$

$$\iint_{ABC} \boldsymbol{F}\cdot n dA + \iint_{ABD} \boldsymbol{F}\cdot n dA + \iint_{ACD} \boldsymbol{F}\cdot n dA + \iint_{BCD} \boldsymbol{F}\cdot n dA = 0$$